

IS6400 – Business Data Analytics - Group Project Guideline

(Project highlights briefing 5%, Presentation 15%, Final Report 10%: Total = 30%)

The objective for this group project is to give students opportunities to apply data analysis techniques learned in the class to real-world problems (such as in finance, marketing, customer relationship management, competitor monitoring, etc.). You can examine problem encountered within your team, department, student union, company, and family. You should investigate the problem carefully and introduce its significance to your classmates. Problems with an application background are more preferred than problems copied from a textbook.

In general, you may choose to do one of the following two types of projects.

1. The first type of projects is the ones with direct applications in business. Due to the limitation of practice, the amount of data related to this problem can be relatively small and the analysis methods need not be complicated. However, the project solution should be able to be directly used in a business (references from managers are optional but highly preferred). Through this type of projects, I expect you to build some gold-standard solutions for the problems you may meet in future.
2. The second type of projects has a larger amount of data and requires relatively more complicated techniques to address. It is related to business but you may not have a real “customer” in hand. Through this type of projects, you are expected to develop skills on the learned techniques and better prepare for the coming big data trend to your focused industry.

1. Team Registration

Please form a project team (**with max 6 students**) **within your section** and:

GO TO: Canvas – People – Group to add your group members.

2. Group Project Proposal Submission 5%

Begin: End of Week 6 Due: End of **Week 7**

At this stage, you should have already obtained the dataset or at least have located the data and be ready to get it. You should have developed a general project plan and had a good idea on how to connect your data with your business problem.

Each group must submit a Group Project Proposal by the end of Week 7.

- Length: No more than 5 pages of PPT

- Required content:

1. Business Problem – Clearly identify the business problem your project aims to address.
2. Dataset Description – Specify the dataset you plan to use, including feature names and data types.
3. Key Variables – Indicate the variables you intend to analyze to solve the problem.
4. BDA Methods – Outline the analytical or modeling methods you plan to apply.
5. Potential Contribution – Highlight the expected business insights or contributions.

Note: To help you prepare for the Week 7 proposal submission, detailed instructions, guidance, and training will be provided in **Week 6 (lecture + tutorial)**.

Preview: Week 6 will be an engaging and meaningful session designed to equip you for success!!!

3. Final presentation (13 mins + 2min Q&A)

15%

Week 12 - 13

In the final presentation, you need to present the entire project to the class, including the business problem, data, your analysis, findings and conclusions.

Please pretend that your audiences are managers in your company, and explain your project and major findings to your managers. (At this stage, major findings should be finished. However, you can still have some minor changes on analysis and findings in the final report).

- (1) Each team will have a limited time for the briefing, so be prepared to highlight the main analysis and findings.
- (2) **All team members** need to present in this session. **All team members** need to attend the Q&A session.

Note: The team correspondent needs to upload the **presentation slides** to canvas before the submission deadline.

4. Project report

10%

Due by May 1, 11:59PM

In general, the final report is expected to be in the range of 20 – 25 pages (11 fonts, double space) in order to cover the required components. (Please include tables and figures in the main text, put references in the appendix, which is counted in the page limit)

Your report should address the following aspects:

- A Cover Page with team members info;
- An one-page Executive Summary;
- Description of the specific business problem;
- Description of the data you collected (and data collection process if any);
- Description of your analytical procedure and related analytical results, for example, you first apply descriptive statistics to show XXX, then you apply regression analysis to show YYY, after that, you conduct time series (or text analytics) to find ZZZ. Please interpret your results in the way that a layman can understand.
- Conclusion;
- References (e.g., journal articles, textbooks, online magazines, etc.); (The references may not need to be comprehensive since this is not an academic paper).
- Any other materials that you think are relevant and interesting.

Note: The team correspondent needs to upload the **soft copy of your report** to Canvas before the deadline. NO HARD COPY IS NEEDED.